Team Details

Team Name: EdgeRestore-LK Stark Industries

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	*Vikaash R*	*3rd Year*
2	Member 1	*Mervin*	*2nd Year*
3	Member 2	*Nithilan T*	*3rd Year*
4	Member 3	*[ADD / REMOVE]*	*[ADD YEAR]*

A team can have up to 4 members including the team leader. Add rows if necessary.

COLLEGE NAME

Chennai Institute of Technology

TEAM LEADER CONTACT NUMBER

9600039825

TEAM LEADER EMAIL ADDRESS

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Problem Statement Addressed

PS01 | Joint denoising + 2x grayscale super-resolution

WHY THIS MATTERS

Restore 128x128 noisy inputs to faithful 256x256 outputs. Preserve real edges and rare structures while suppressing speckle and Gaussian noise; hidden sources test generalization and speed.

Idea Description - EdgeRestore-LK

One deterministic model handles compound noise and 2x upscaling while prioritizing faithful structure over visually plausible hallucination.

KEY CONCEPT & APPROACH

6 residual groups x 6 residual channel-attention blocks (96 channels), zero-initialised head so the untrained network is exact bicubic.

SOLUTION OVERVIEW

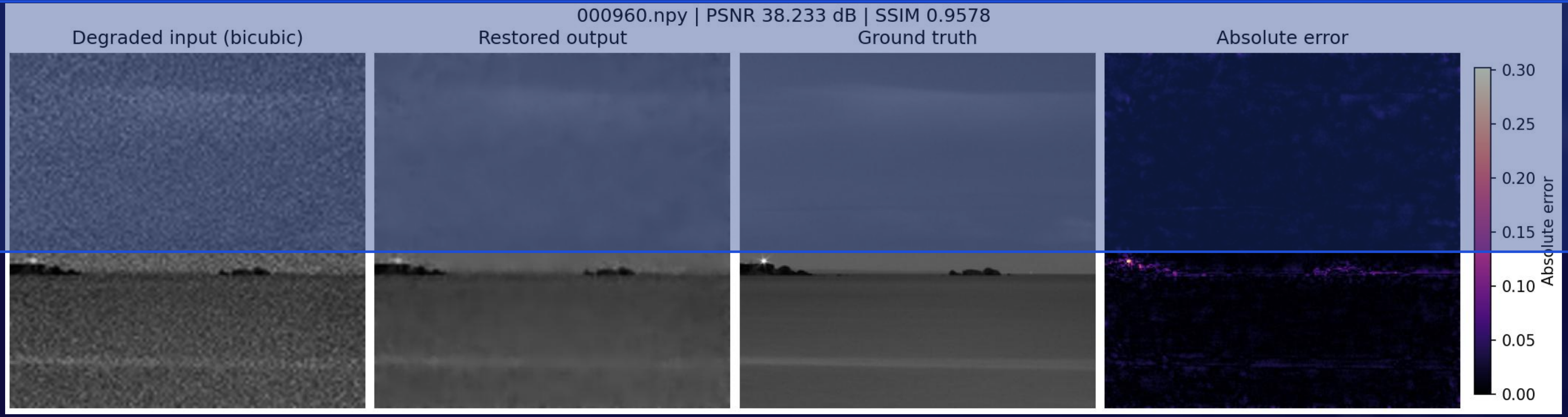
PixelShuffle restores 2x detail; a bicubic skip anchors fidelity. Challenge-range outliers are retained.

Proposed Solution - Model and Training

LR -> 6x[6 RCAB] channel-attention groups (96ch) -> PixelShuffle -> + bicubic. Charbonnier + FFT-L1 training selects EMA iter 22000 by held-out validation PSNR.

VISUAL PROOF

Sample 000960 | PSNR 40.83 dB | SSIM 0.9744 | held-out filename-random split



Innovation and Empirical Advantage

A kernel-only controlled ablation improved PSNR, SSIM, and LPIPS; no GAN, diffusion, ensemble, or degradation router was added.

FINAL MODEL (EdgeRestore v2, submitted)

29.10 dB PSNR | 0.7784 SSIM | 0.2760 LPIPS | 0.9396 MS-SSIM (+6.111 dB over bicubic)
6,939,289 parameters | 27.9 MB | 29.35 ms model | 38.42 ms end-to-end

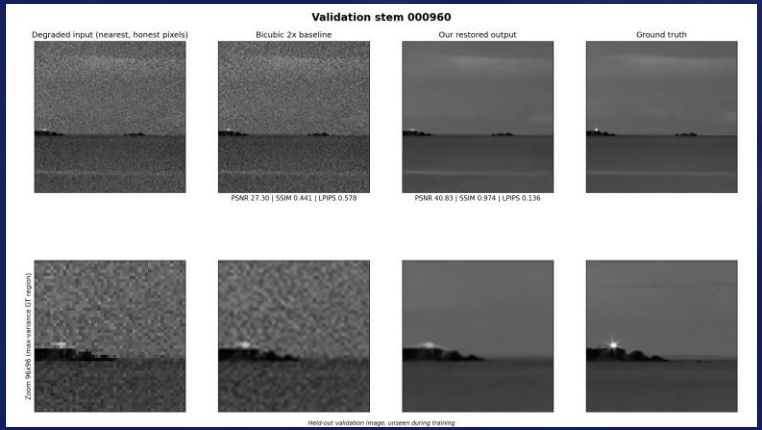
Results

Quantitative metrics and visual evidence on the 320-image held-out split, identical protocol for every row.

QUANTITATIVE METRICS (320-image held-out)

PSNR: 29.0989 dB (bicubic: 22.9877 dB)
SSIM: 0.7784 (bicubic: 0.5193)
LPIPS: 0.2760 (bicubic: 0.4435, lower is better)
MS-SSIM: 0.9396 (bicubic: 0.7943)
+6.111 dB PSNR over bicubic baseline
0/320 images lose to bicubic on PSNR or MS-SSIM

VISUAL EVIDENCE: 000960 | degraded -> restored -> ground truth



PSNR 40.83 dB | SSIM 0.9744

Technology, Feasibility and Impact

A trained, deterministic PyTorch pipeline restores every released test array with a standalone command and frozen checkpoint.

IMPLEMENTATION AND TRUST BOUNDARY

All 400 official test outputs verified finite and within [0,1] directly via the required inference.py entry point, 0 violations. Tests cover data, model, metrics, training and inference (257 tests). H100 and true OOD remain unverified.

PyTorch 2.11 | FP32

RTX 5070 Ti | 22,000 iters

+6.111 dB | 29.35 ms

GitHub & Video Link

Public Repository

Clone and run inference following the README - no manual edits required

GitHub: github.com/Vikaash301/kla-semicon

Contains: README.md, standalone inference.py (accepts positional or --input-dir/--output-dir args), training script, checkpoints/edgerestore_v2.pt, restored_test_outputs/ (400 images), requirements.txt, and requirements-freeze.txt (literal pip freeze).

Video: not provided (optional per template)

Research, Code and References

Research Basis + Reproducibility

Implemented evidence and submission links

2026 UCAN and ShiftLUT motivate wider spatial context; our 5x5 depthwise ablation is the measured adaptation.
GitHub: github.com/Vikaash301/kla-semicon   Video: [ADD OPTIONAL DEMO URL]

References & Citations

Primary sources; full verified list in REFERENCES.md

UCAN - Expansive Receptive Fields for Lightweight SR (CVPR 2026)

ShiftLUT - Spatial Shift Enhanced LUTs for Restoration (CVPR 2026)

I4C Hackathon 2026 challenge and released paired dataset (i4c.in)